

Dataset EDA — 04_dataset.jsonl

Generated from `data/pipeline/04_dataset.jsonl` (119,040 entities).

Splits

Split	Entities
train	95,173
val	11,893
test	11,974
total	119,040

Splits are deterministic: MD5 hash of `entity_id` modulo 1000, bucketed at 80/10/10.

Positives per Entity

Metric	Value
min	0
mean	39.2
max	72
std	8.4

Each entity has ~39 positive pairs on average: 4 phonetic Latin variants + 5 names (`name_en` + 4 variants) × 8 scripts, minus deduplication.

Positive Type Distribution

Type	Count
combined	3,290,117
script	918,410
phonetic	460,930
total	4,669,457

- **phonetic**: Latin script variants of `name_en` (LLM-generated phonetic perturbations)
 - **script**: `name_en` transliterated into a non-Latin script
 - **combined**: Latin variants (not `name_en`) transliterated into non-Latin scripts
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Source Distribution

Source	Count	%
llm	4,644,259	99.5%
wikidata	25,198	0.5%

The dataset is almost entirely LLM-generated. Wikidata provides ground-truth names for a small subset of entities.

Script Distribution (LLM vs Wikidata)

Script	Total	LLM	LLM %	Wikidata	Wiki %
ar	516,898	515,080	100%	1,818	0%
el	533,170	532,597	100%	573	0%
he	519,858	518,877	100%	981	0%
hi	528,773	528,607	100%	166	0%
ja	529,857	522,324	99%	7,533	1%
ko	521,051	519,566	100%	1,485	0%
ru	541,946	532,883	98%	9,063	2%
zh	516,974	513,395	99%	3,579	1%

Script coverage is nearly uniform. Russian and Japanese have the highest Wikidata coverage (2% and 1% respectively), reflecting their prevalence in Wikidata.

Anchor Name Length (UTF-8 bytes)

Metric	Value
min	2
mean	16.2
max	116
std	6.0

All names are well within the model's `max_len=256` byte limit.